Worksheet 33 · Pythagoras' Theorem IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:

Learning Objectives:
Students will be able to:

- Find the hypotenuse of a right-angled triangle
- Find a shorter side of a right-angled triangle
- Apply Pythagoras' theorem to real-life problems

Grade: / 53	Teacher's Feedback:

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Pythagoras' Theorem*

Question 1 (3 marks)

The diagram shows a rectangle and a diagonal of the rectangle.

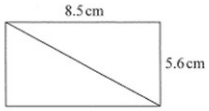


Diagram **NOT** accurately drawn

Work out the length of the diagonal of the rectangle.
Give your answer correct to 1 decimal place.

..... cm

Question 2 (3 marks)

Here is a right-angled triangle.

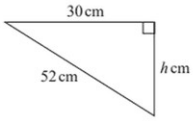


Diagram **NOT** accurately drawn

Calculate the value of h.
Give your answer correct to 3 significant figures.

h = ...

Question 3 (4 marks)

Two circles, C_1 and C_2, are drawn on a centimetre grid, with a scale of 1 cm for 1 unit on each axis.

The centre of circle C_1 is at the point with coordinates (–1, 3) and the radius of C_1 is 13 cm.

The centre of circle C_2 is at the point with coordinates (7, 18) and the radius of C_2 is 6 cm.

(a) Work out the distance between the centre of C_1 and the centre of C_2

(b) Explain why circle C_1 intersects circle C_2

.....

.....

(1)

Question 4 (4 marks)

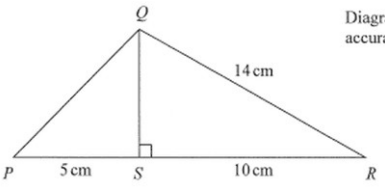


Diagram **NOT** accurately drawn

In triangle PQR,

S is the point on PR such that angle $RSQ = 90°$
$RQ = 14$ cm
$RS = 10$ cm
$SP = 5$ cm

Work out the length of PQ.

Question 5 (4 marks)

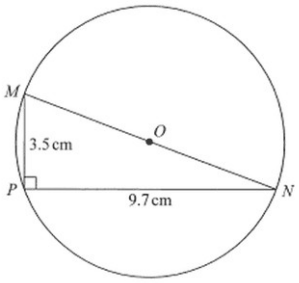


Diagram **NOT** accurately drawn

M, N and P are points on a circle, centre O.
MON is a diameter of the circle.

$MP = 3.5$ cm
$PN = 9.7$ cm

Angle $MPN = 90°$

Work out the circumference of the circle.
Give your answer correct to 3 significant figures.

Question 6 (3 marks)

The diagram shows isosceles triangle ABC.

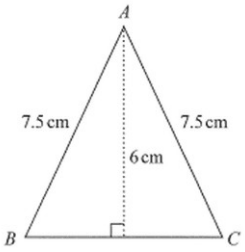


Diagram **NOT** accurately drawn

$AB = AC = 7.5$ cm.
The height of the triangle is 6 cm.

Calculate the area of the triangle.

Question 7 (4 marks)

The diagram shows a shape made from a right-angled triangle and a semicircle.

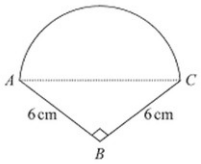


Diagram **NOT** accurately drawn

AC is the diameter of the semicircle.
$BA = BC = 6$ cm
Angle $ABC = 90°$

Work out the area of the shape.
Give your answer correct to 1 decimal place.

Question 8 (4 marks)

The diagram shows an isosceles triangle.

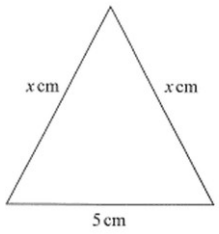


Diagram **NOT** accurately drawn

The area of the triangle is 12 cm^2

Work out the perimeter of the triangle.
Give your answer correct to 3 significant figures.

Question 9 (4 marks)

The shaded shape is made using three identical right-angled triangles and a square.

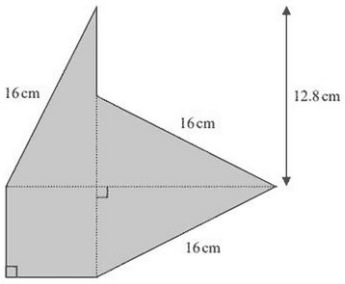


Diagram **NOT** accurately drawn

Work out the perimeter of the shaded shape.

Question 10 (5 marks)

A, B and C are points on a circle with centre O.

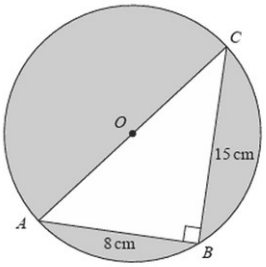


Diagram **NOT** accurately drawn

AOC is a diameter of the circle.

$AB = 8$ cm $BC = 15$ cm

Angle $ABC = 90°$

Work out the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

Question 11 (3 marks)

The diagram shows isosceles triangle ABC

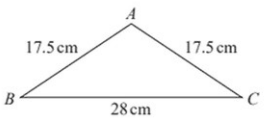


Diagram **NOT** accurately drawn

$AB = AC = 17.5$ cm $BC = 28$ cm

Calculate the area of triangle ABC

Question 12 (4 marks)

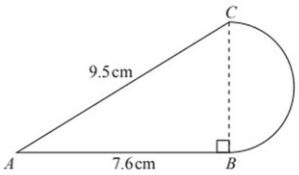


Diagram **NOT** accurately drawn

The diagram shows a shape made from triangle ABC and a semicircle with diameter BC.
Triangle ABC is right-angled at B.

$AB = 7.6\,\text{cm}$ and $AC = 9.5\,\text{cm}$.

Calculate the area of the shape.
Give your answer correct to 3 significant figures.

Question 13 (4 marks)

Here is a triangular prism.

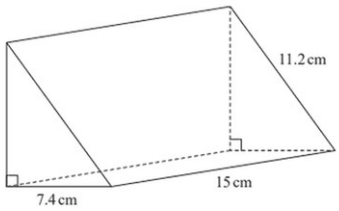


Diagram **NOT** accurately drawn

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

Question 14 (4 marks)

The diagram shows a quadrilateral $ABCD$

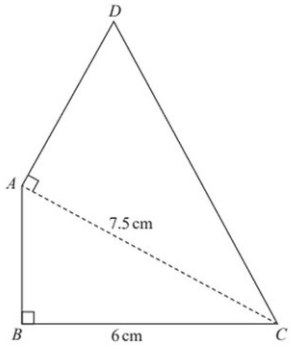


Diagram **NOT** accurately drawn

In the diagram, ABC and DAC are right-angled triangles.

$BC = 6\,\text{cm}$ $AC = 7.5\,\text{cm}$

The area of quadrilateral $ABCD$ is $31.5\,\text{cm}^2$

Work out the length of AD